Sprint 6 – Missile Shoot with State and Collision

Student Information

Integrity Policy: All university integrity and class syllabus policies have been followed. I have neither given, nor received, nor have I tolerated others' use of unauthorized aid.

I understand and followed these policies: ◯ Yes ◯ No

Name:

Date:

Submission Details

Final ***Changelist*** number:

Verified build: ◯ Yes ◯ No

Required Configurations: Debug only

YouTubeLink:

Discussion (What did you learn):

Verify Builds

- Follow the Piazza procedure on submission
  - Verify your submission compiles and works at the changelist number.
- Verify that only MINIMUM files are submitted
  - No – Generated files
    - *.pdb, *.suo, *.sdf, *.user, *.obj, *.exe, *.log, *.pdb, *.db, *.user
    - Anything that is generated by the compiler should not be included
  - No – Generated directories
    - /Debug, /Release, /Log, /ipch, /.vs
- Typical files project files that are required
  - *.sln, *.csproj, *.cs,
  - App.config, AssemblyInfo.cs, CleanMe.bat
  - Resources Directory:
    - *.tga, *.dll, *.wav, *.glsl, *.azul

Standard Rules

Submit multiple times to Perforce

- Submit your work as you go to perforce several times
  - Design Patterns – no minimum
  - Sprints – minimum of 5 real submissions (generally 10+)
    - As soon as you get something working, submit to perforce
    - Have reasonable check-in comments
      - Points will be deducted if minimum is not reached

Submission Report

- Fill out the submission Report
  - No report, no grade

Code and project needs to compile and run

- Make sure that your program compiles and runs
  - Warning level 4
  - NO Warnings or ERRORS
    - Your code should be squeaky clean.
  - Code needs to work “as-is”.
    - No modifications to files or deleting files necessary to compile or run.
  - All your code must compile from perforce with no modifications.
    - Otherwise it's a 0, no exceptions

Project needs to run to completion

- If it crashes for any reason…
  - It will not be graded and you get a 0

No Containers

- Containers (No automatic containers or arrays
- Template or generic parameters
- No arrays
  - You need to do this the old fashion way - ***YOU EARNED IT***

Leave Project Settings

- Do NOT change the project or warning level
  - Any changing of level or suppression of warnings is an integrity issue

Simple C#

- No .Net
- We are using the basics
  - Types:
    - Class, Structs, intrinsic types (int, float, bool, etc…)
    - NO arrays allowed!
  - Basics language features
    - Inheritance, methods, abstract, virtual, etc…

No Debug code or files disabled

- Make sure the program has only active code
  - If you added debug code or commented out code,
    - please return to code to active state or remove it

Adding files to this project

- Make sure you add the files in the appropriate sub-directories
- Make sure any new files are successfully integrated into the project
- Make sure your new files are submitted to Perforce

Due Dates

- See Piazza for due date and time
- Submit program perforce in your student directory assignment supplied.
- Fill out your this ***Submission Report*** and commit to perforce
  - ***ONLY*** use Adobe Reader to fill out form, all others will be rejected.
  - Fill out the form and discussion for full credit.

Goals

- Ship movement state – MoveRight, MoveLeft, MoveBoth
- Missile state – Flying, Ready
- Bumpers – for ship movement (right/left extreme)
- Collision between Missiles and Alien Grid
- Collision between missile vs top wall
- Collision between Ship vs Bumper
- Movement of delta time and delta_x changes proportional to number of aliens

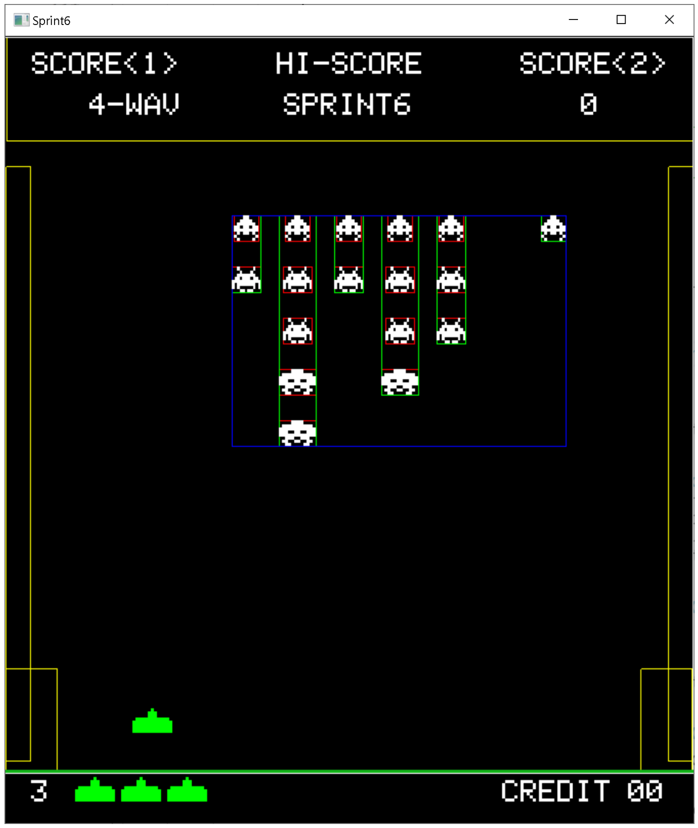
Assignments

Grading:

Summary of features: https://youtu.be/q2yePm9t-7U

- ***Ship movement with Bumpers***
  - Ships are limited by the right/left bumper through collision and observer callback
- ***Ship State***
  - Movement controlled by state
    - MoveRight,MoveLeft,MoveBoth
- ***Missile State***
  - Can only shoot one missile at a time
  - States – Flying and Ready
  - Changes state from flying when the missile terminates through collision
    - Top Wall or Aliens
- ***Missile vs Alien Grid***
  - Delete alien and missile to clear the grid
  - Deletes both the Game Object and corresponding Collision Object with collision box
    - No cheating by painting box Black
  - Changes Missile state from Flying to Ready
- ***Missile vs Top Wall***
  - Deletes Missile and Changes missile state from Flying to Ready
- ***Alien March Speed up***
  - Aliens start to move horizontally with 4 pixels per update
    - As the number of aliens goes from 55 to 0…
    - Horizontal Movement increase from 4 to 16 proportional to the number of aliens
  - Aliens start delta time (Drum Beat) is the time between movement and marching sounds
    - As the number of aliens goes from 55 to 0…
    - Delta time decreases from 0.7sec to 0.05sec portional to the number of aliens

Required Features:



Sizes and values need for this demo

- Bumpers:
  - 50x100 (w x h) on the right/left wall
  - Center at y=100 on the right/left side
- Floor (bottom wall):
  - 672x3 (w x h)
  - starting y=50
- Ceiling (top wall):
  - 672x103 (w x h)
- Ship:
  - 39x24(w x h)
  - (336,100) (x,y) start
  - Speed of movement: delta_x = 2.0f
- Missile:
  - 3x12 (w x h)
  - Starts at (ship.x,ship.y+20)
  - Speed of movement: delta_y = 9.0f
- Reserve ships
  - Reserve Ships (x,y) same as main ship 39x24(w x h)
    - (88, 33) (x,y)
    - (133, 33) (x,y)
    - (178, 33) (x,y)

- Bottom Fonts
    - Fonts are 24 pixels tall,and 15 wide.... exception "space" which is 3 pixels wide
        - (25, 30) (x,y) start // 3
        - (485, 30) (x,y) start // Credits
        - (608, 30) (x,y) start // 00
- Grid Movement
    - Movement proportional to number of aliens
    - As alien count decreases... the delta_time decreases and delta_x increases
    - Grid movement by delta_time
        - Start delta_time: 0.7s at 55 aliens
        - End delta_time: 0.05s at 0 aliens
    - Grid horizontal movement in x direction
        - Start delta_x: 4 pixels per update at 55 aliens
        - End delta_x: 16 pixels per update at 0 aliens

States - as described above affect the ship/missile

- Movement State
- Missile state

Demo – show in Video

- Needs to show states (movement and missile state)
- Need to show bumpers
- Need to show missile vs ceiling
- Need to show missile vs alien grid
- Need to show grid speed up proportional
- Need to show the bottom reserve ships and fonts

Approach

- Start with Sprint5 and modify and extend
- Reduce the problem to allow you to focus on a feature one at a time
- Mimic screen shot and video

Video Capture

- Have Visual Studio open and ready to go.
- Start recording
    - Show output window running
    - Speak clearly and loud
    - Discuss the features you added in this demo
    - Duration
        - Approx. 5-10 min long YouTube video capture
            - Do not exceed 10 min
        - Use Zoom recording
            - High resolution
            - Good sound quality

        - Show on the capture… it compiling and then running
    - Place YouTube link in PDF
    - Make sure it public and runs without passwords or login
    - Watch your video
        - Can you see the code clearly
        - Can you hear your voice clearly

Make sure you run CleanMe.bat before perforce submission

Validation

Simple checklist to make sure that everything is submitted correctly

- Is the project compiling and running without any errors or warnings?
- Does the project runs ***ALL*** without crashing?
- Is the submission report filled in and submitted to perforce?
- ***Follow the verification process*** for perforce
    - Is all the code there and compiles "as-is"?
    - No extra files
- ***Submit the YouTube link to PDF***

Hints

Most assignments will have hints in a section like this.

- Watch the lecture… look at the code supplied in class
- Study the projects in class lecture
- ASK questions… a lot of moving parts in this sprint

Troubleshooting

- Print to the output window to help you debug
    - It really helps
- Ask for clarification on piazza